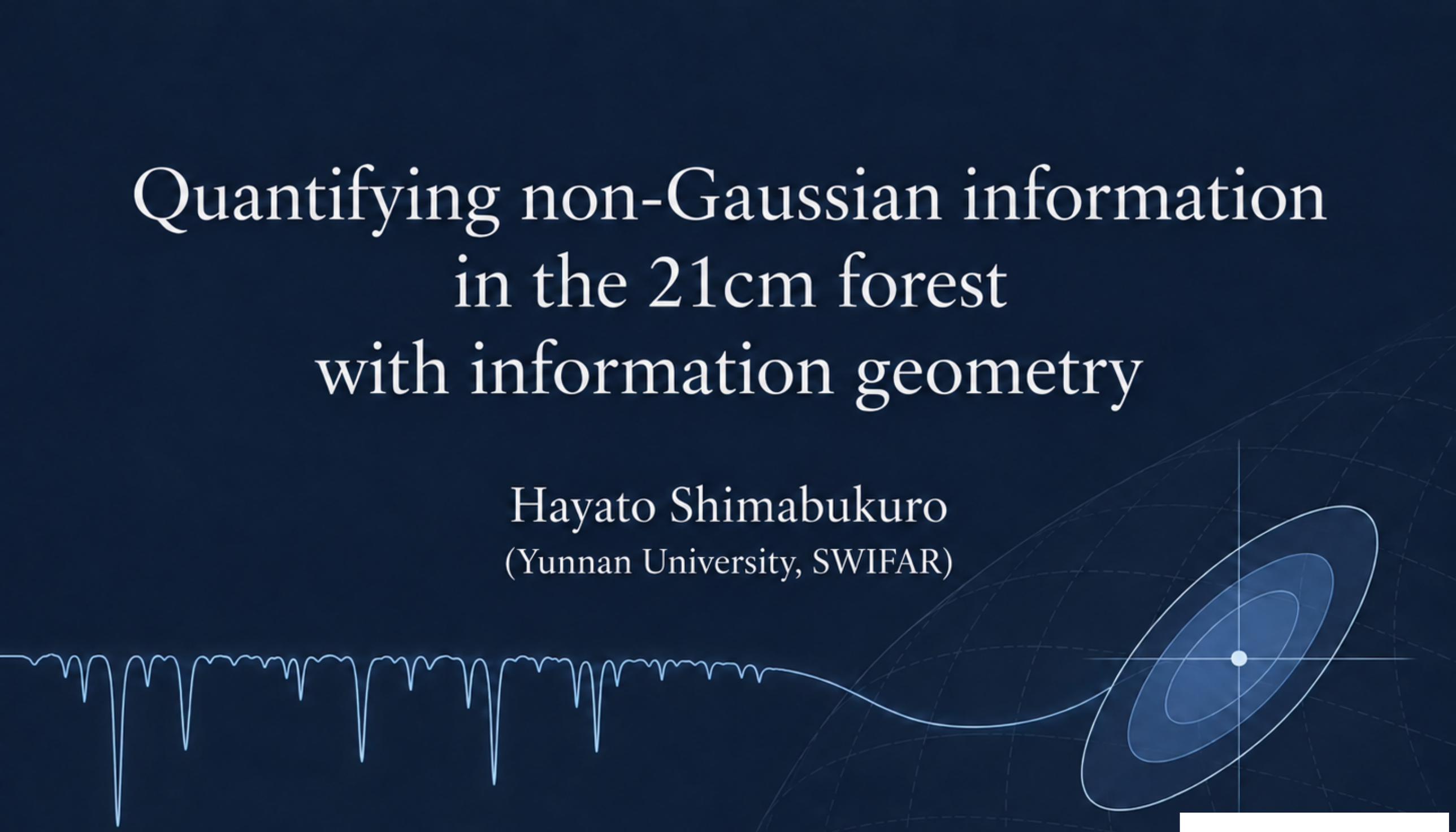
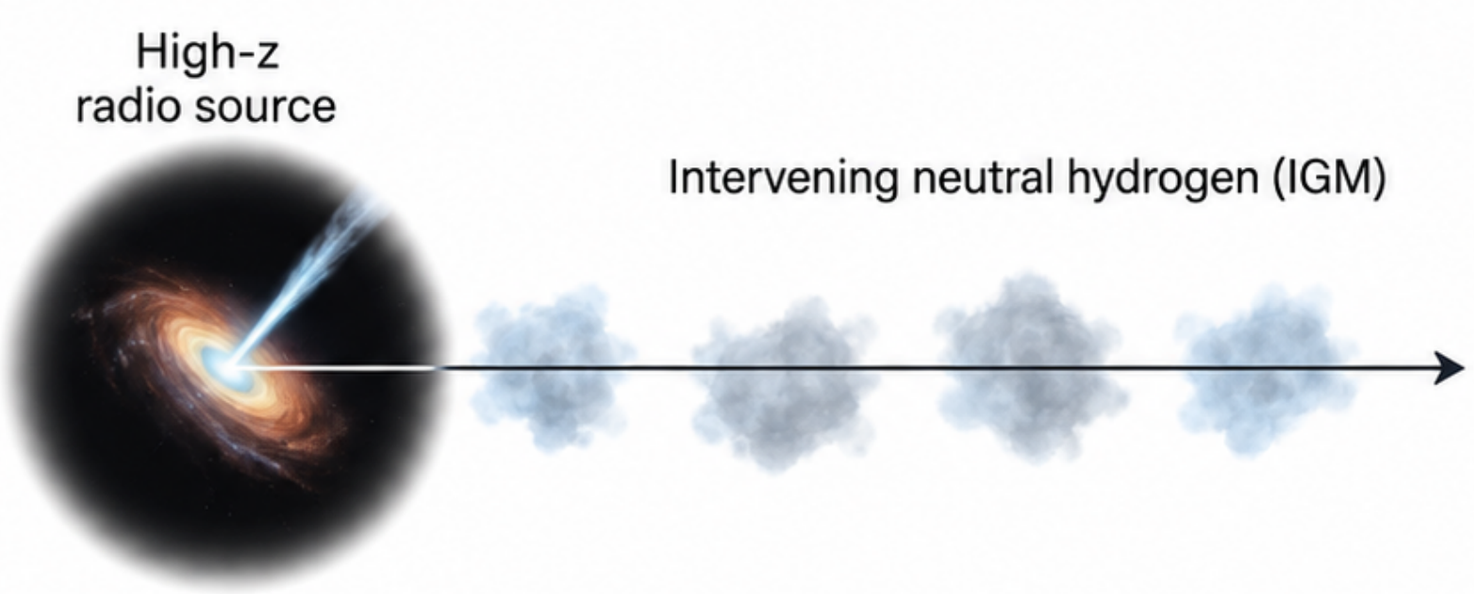
Quantifying non-Gaussian information in the 21cm forest with information geometry

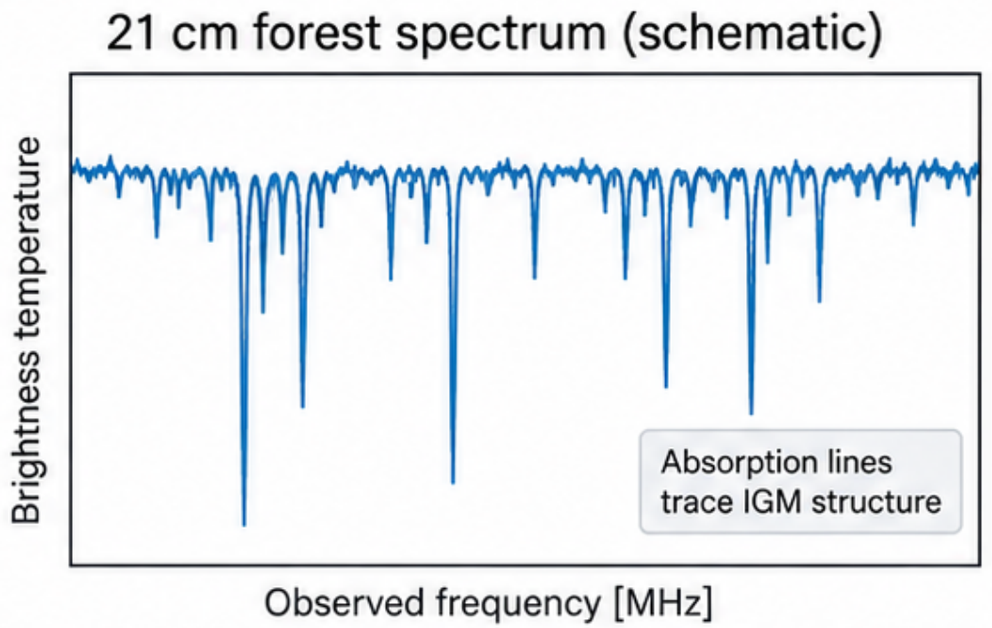
Hayato Shimabukuro
(Yunnan University, SWIFAR)



This paper has been “on-hold” on arXiv for 2 weeks…

Why non-Gaussian information in the 21 cm forest?





Rich small-scale structure

Absorption features encode the density, temperature, and velocity field of the IGM.

Beyond the power spectrum

The power spectrum uses Fourier amplitudes only and ignores phase information.

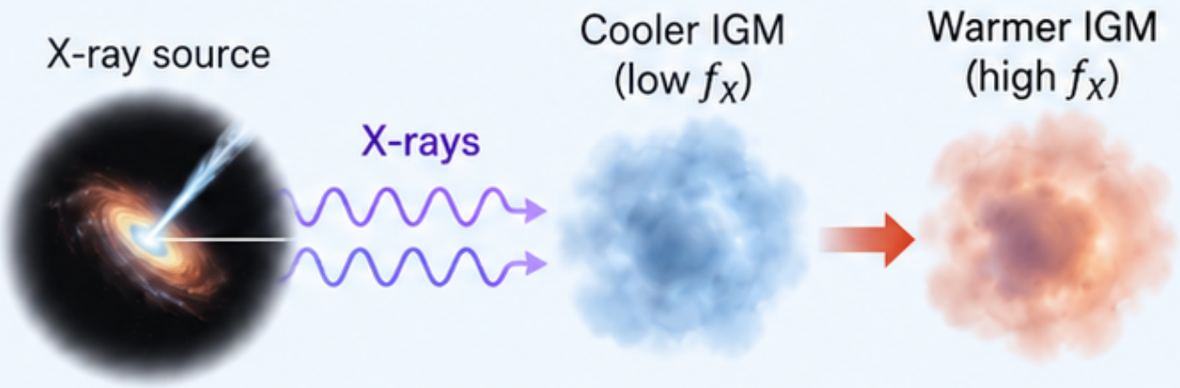
Potential in morphology

Non-Gaussian structure and Fourier phase may contain additional information relevant for parameter inference.

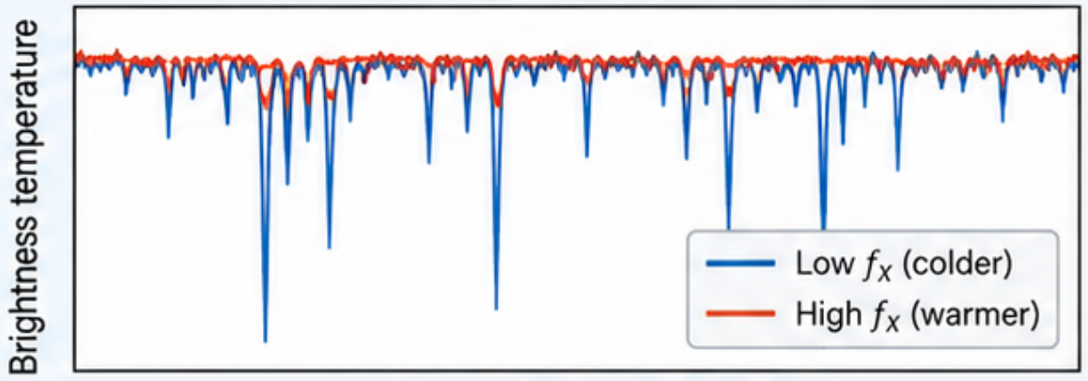
Does non-Gaussian morphology improve physical parameter inference?

Two physical parameters

X-ray heating: f_X

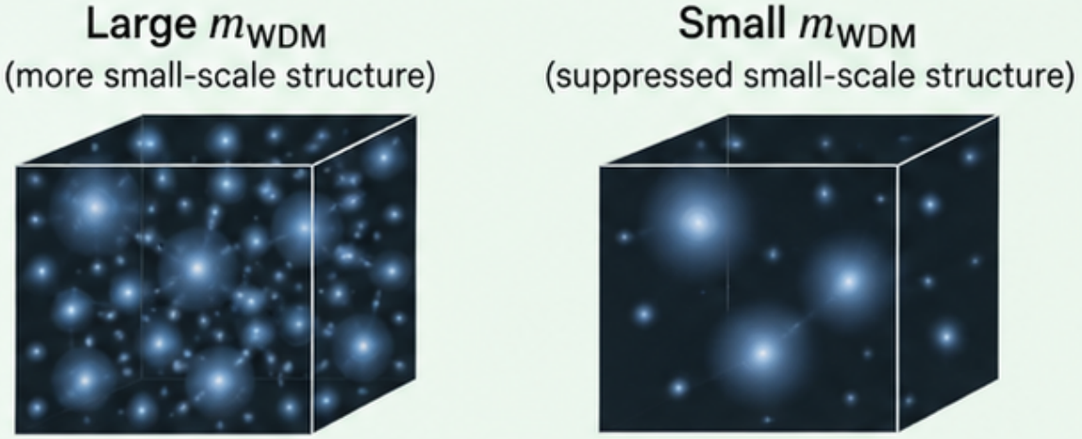


21 cm forest spectrum (schematic)

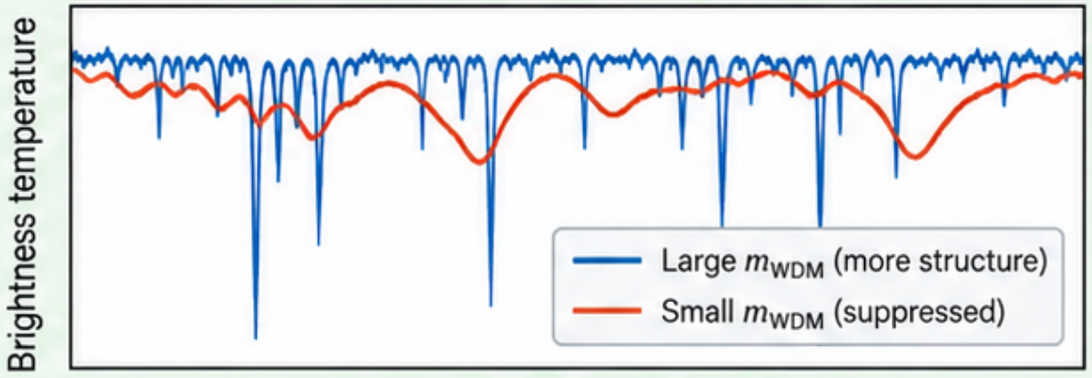


- Controls the thermal state of the IGM
- Higher f_X → warmer gas
- Absorption becomes weaker

Warm dark matter: m_{WDM}



21 cm forest spectrum (schematic)



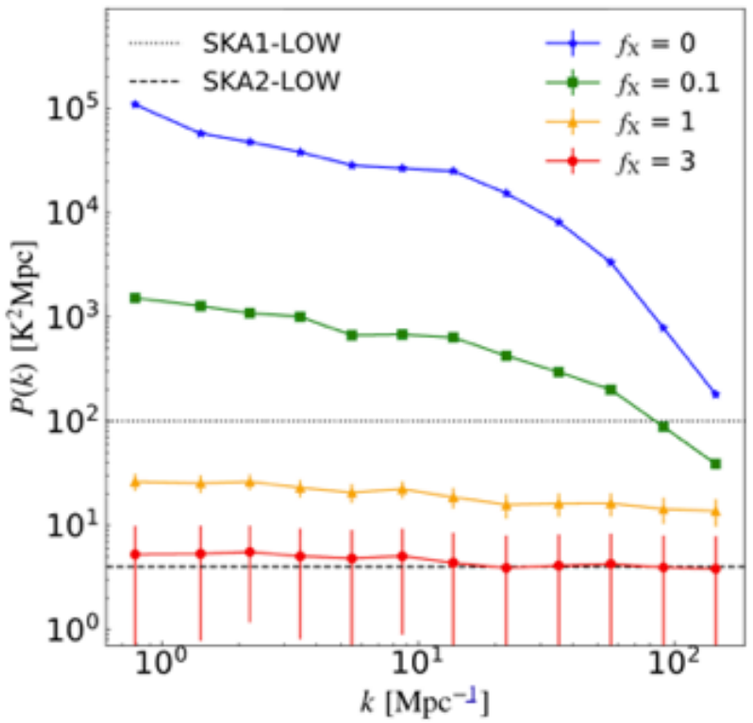
- Controls small-scale structure
- Lower m_{WDM} → stronger free streaming
- Narrow absorption features change

Different physics, different responses in the 21 cm forest.

Three complementary summaries of the 21 cm forest

Power spectrum (PS)

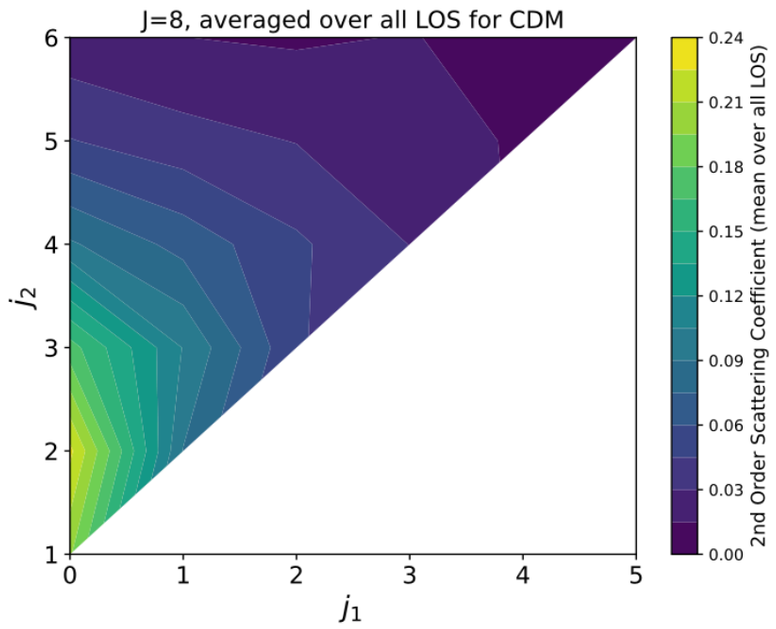
Statistics of fluctuations in Fourier space



Fourier amplitudes → power vs. scale

Wavelet scattering transform (WST)

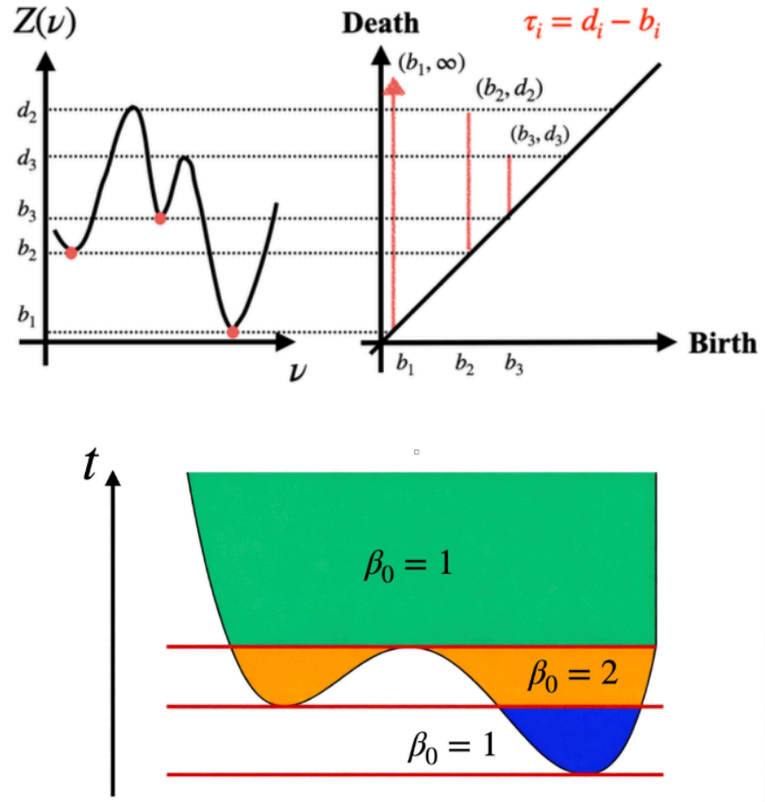
Multiscale, non-linear statistics



S_1: amplitudes S_2: cross-scale modulation

Topological data analysis (TDA)

Morphology and connected structure



Persistence diagram / Betti-0 curve

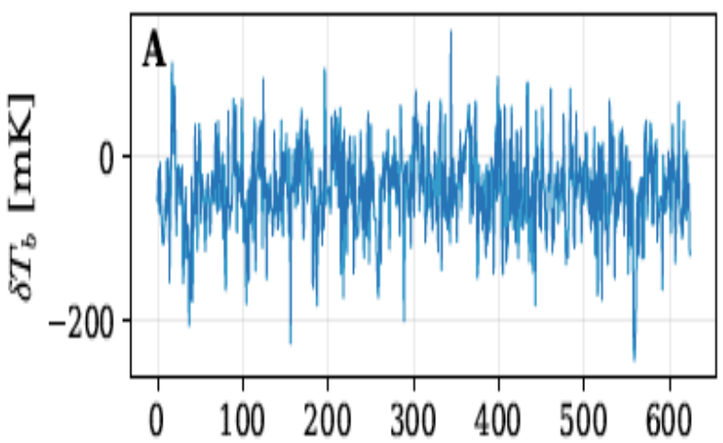
Same forest, different summaries: amplitudes, multiscale coupling, and topology

Shao et al., *Nat. Astron.* **7**, 1116–1126 (2023) | Shimabukuro, Xu & Shao, *Phys. Rev. D* **112**, 063557 (2025) | Shimabukuro, *Phys. Rev. D* **113**, 083525 (2026)

Destroy phases, keep power: a useful diagnostic

Original spectrum

Simulated 21 cm forest

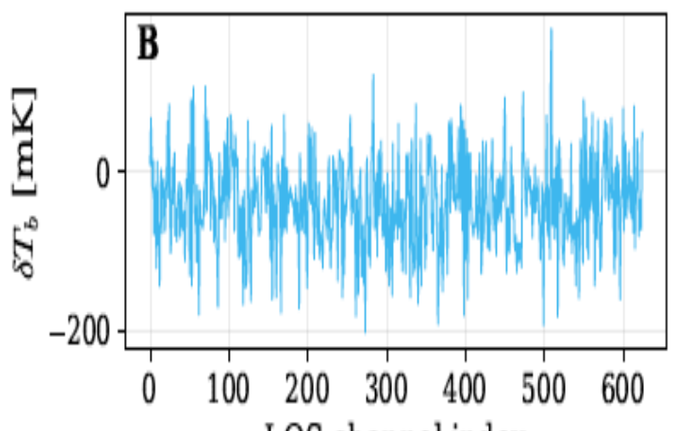


$$\delta T_b(\nu)$$

- Simulated 21 cm forest spectrum with absorption features.
- Contains both amplitude and phase information.

Random-phase surrogate

Keep amplitudes, randomize phases

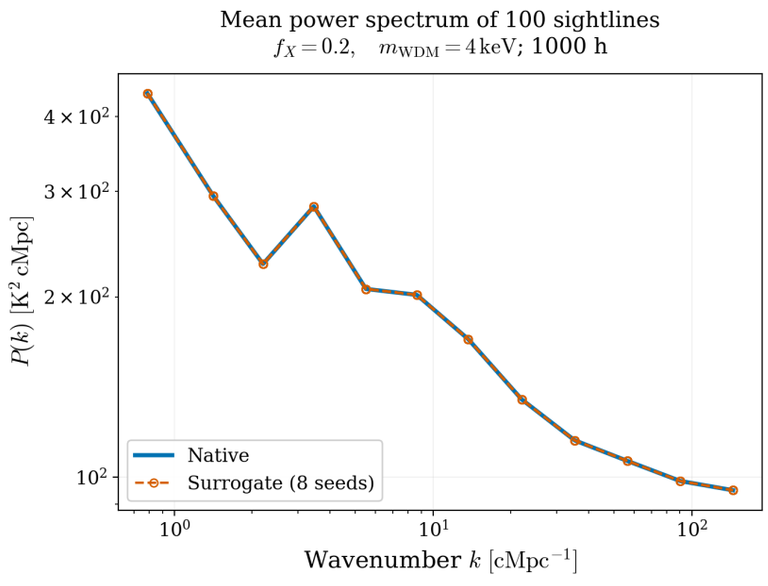


$$\widetilde{\delta T_b}(k) = |\delta T_b(k)| e^{i\phi_k}$$

- Keep Fourier amplitudes $|\delta T_b(k)|$ fixed.
- Randomize Fourier phases $\phi_k \sim \mathcal{U}(0, 2\pi)$.

Power spectrum

Almost unchanged



$$P(k) = \left\langle |\delta T_b(k)|^2 \right\rangle$$

- Power spectra of the original and random-phase spectra are nearly identical.
- Fourier amplitudes alone cannot capture the absorption morphology.

Same Fourier amplitudes, different morphology.

Information geometry: turn distinguishability into geometry

Local parameter space

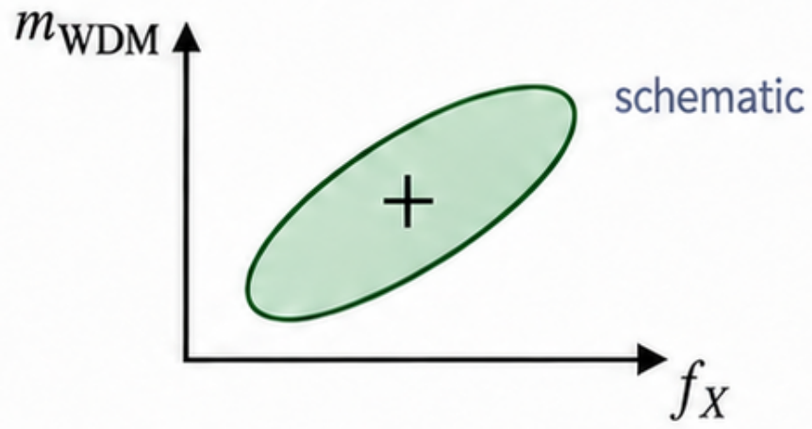
$$\theta = (f_X, m_{\mathrm{WDM}})$$

$$D_{\mathrm{KL}}(\theta, \theta + d\theta) \simeq \frac{1}{2} F_{\alpha\beta}\, d\theta_\alpha\, d\theta_\beta$$

KL divergence: A measure of how one probability distribution differs from another.

Fisher volume density

$$\sqrt{\det F}$$



- Larger $\sqrt{\det F}$ → smaller local error ellipse

For each summary statistic

PS
WST
TDA
→
summary $\mu(\theta)$
→
Fisher matrix F

- Compare how much information each statistic carries
- Repeat after phase randomization

Geometry provides one common language for PS, WST, and TDA.

Information geometry: turn distinguishability into geometry

Local parameter space

$$\theta = (f_X, m_{\rm WDM})$$

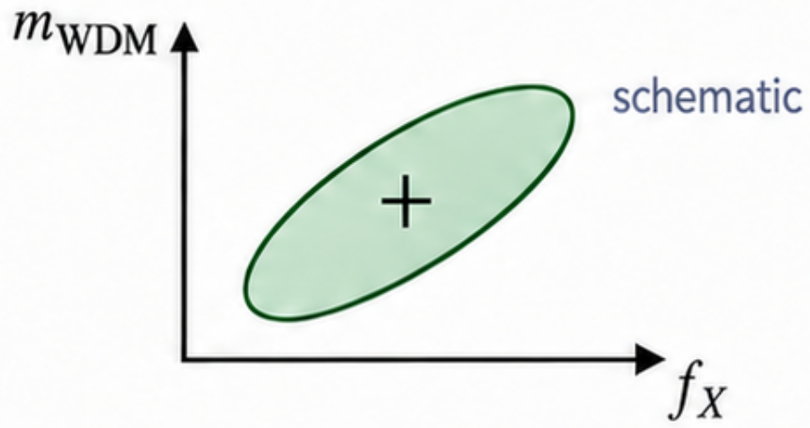
$$D_{\rm KL}(\theta, \theta + d\theta) \simeq \frac{1}{2} F_{\alpha\beta} d\theta_\alpha d\theta_\beta$$

KL divergence: A measure of how one probability distribution differs from another.

- Fisher matrix acts as a local metric

Fisher volume density

$$\sqrt{\det F}$$



- Larger $\sqrt{\det F}$ → smaller local error ellipse

For each summary statistic

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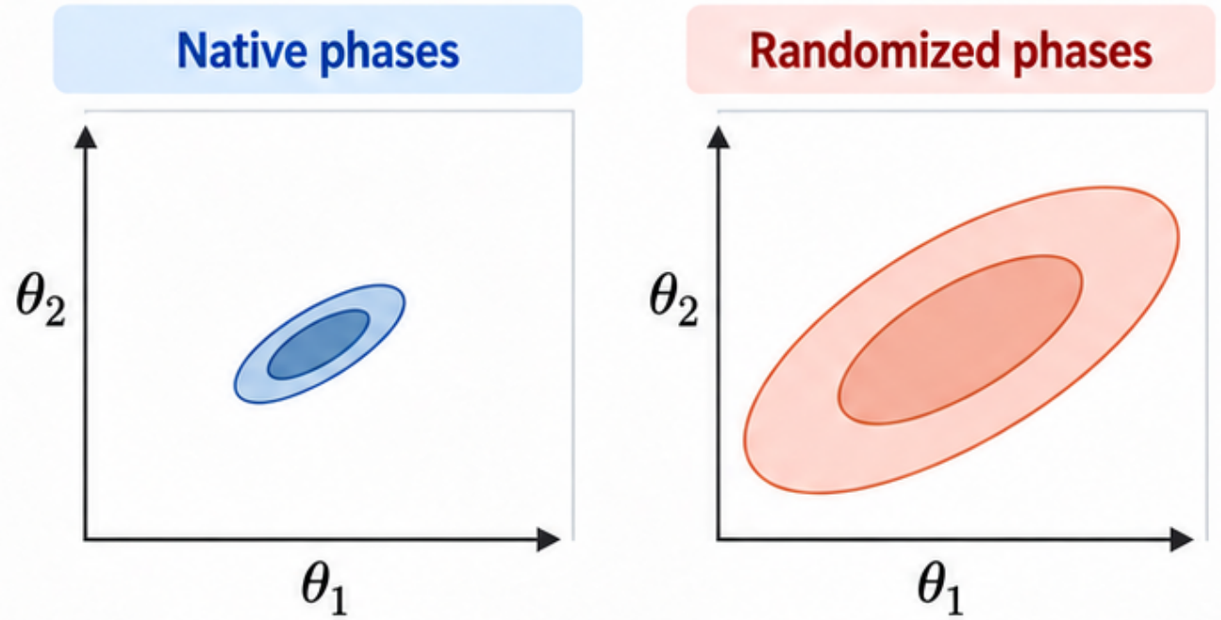
Phase information and information geometry

Define the phase information fraction

$$\eta_\varphi = 1 - \frac{\sqrt{\det F_{\mathrm{surr}}}}{\sqrt{\det F_{\mathrm{native}}}}$$

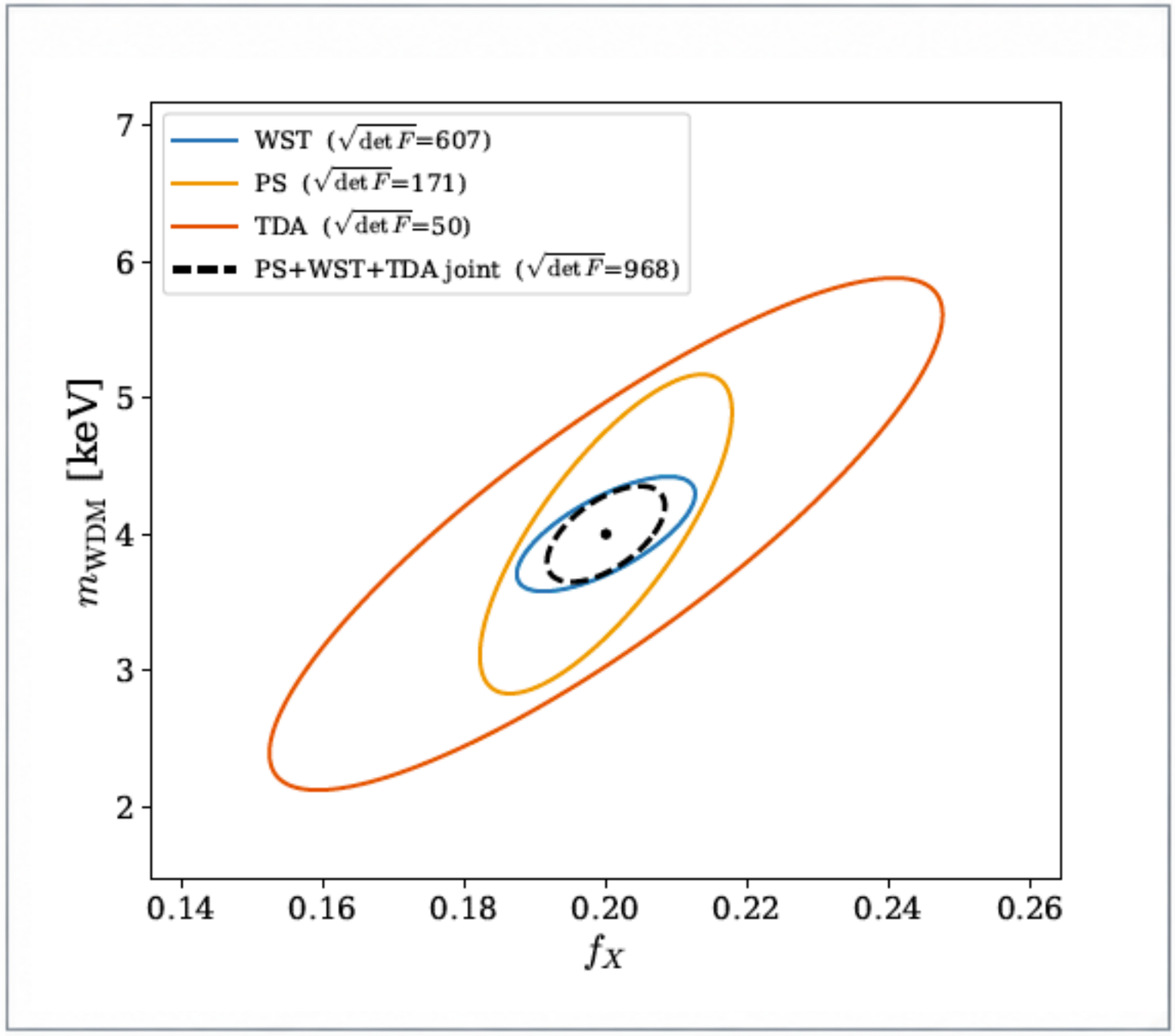
F: Fisher matrix, F_{surr}: randomized phases,
F_{native}: original (native) phases

Fisher constraints (schematic)



- $\eta_\varphi = 0$ — No information is lost by phase randomization.
- $\eta_\varphi > 0$ — Represents the amount of information lost by randomizing the Fourier phases.
- **Non-Gaussian information** — η_φ quantifies the information content of non-Gaussian structures that is destroyed by phase randomization.

WST gives the strongest individual constraint



Benchmark: $f_X = 0.2$, $m_{WDM} = 4$ keV; 1000 h, 10 mJy source

Fisher-volume density

PS: $\sqrt{\det F} = 171$

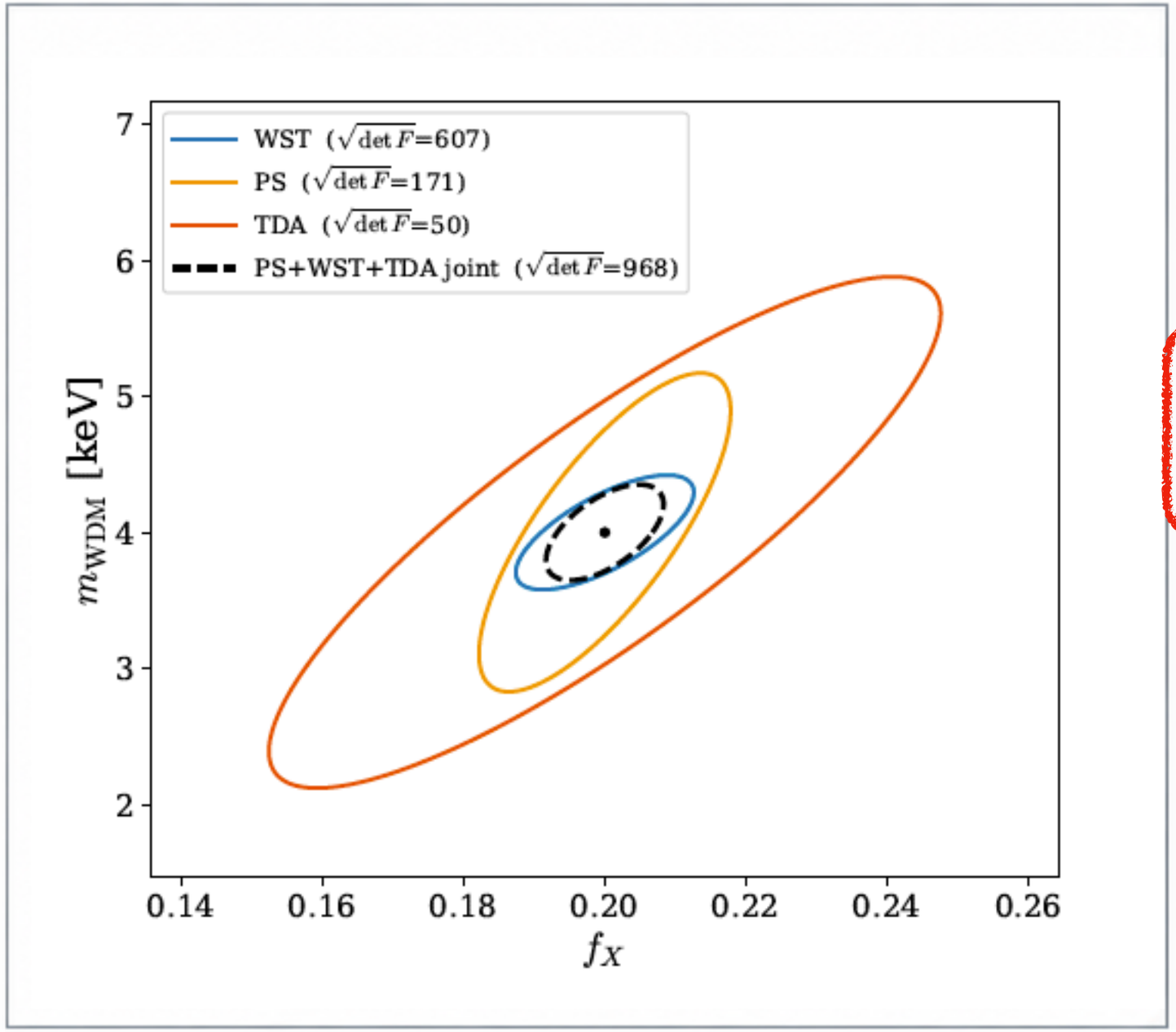
WST: $\sqrt{\det F} = 607$

TDA: $\sqrt{\det F} = 50$

- WST / PS ≈ 3.6 at the benchmark point
- Total constraining power ≠ phase sensitivity

First result: WST carries the largest local Fisher information among the three individual summaries.

WST gives the strongest individual constraint



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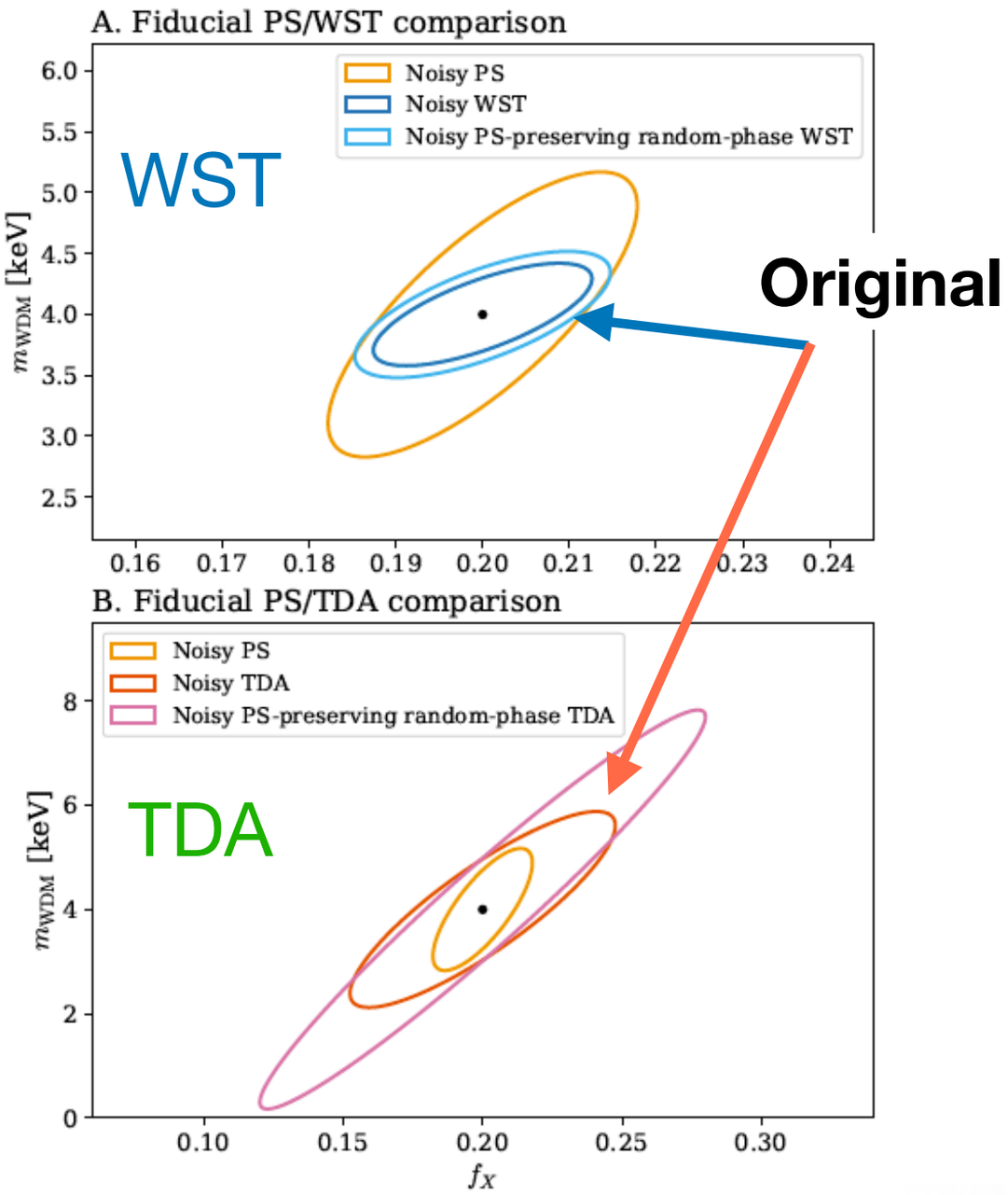
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First result: WST carries the largest local Fisher information among the three individual summaries.

How much survives phase randomization?



Original

WST

$$\sqrt{\det F} = 607.3 \rightarrow 405.1$$

$$\eta_\phi \simeq 0.33$$

- About $2/3$ of the Fisher-volume density remains after phase randomization.
- WST provides stronger constraints than PS, but most of its information survives phase randomization.

TDA

$$\sqrt{\det F} = 49.9 \rightarrow 29.2$$

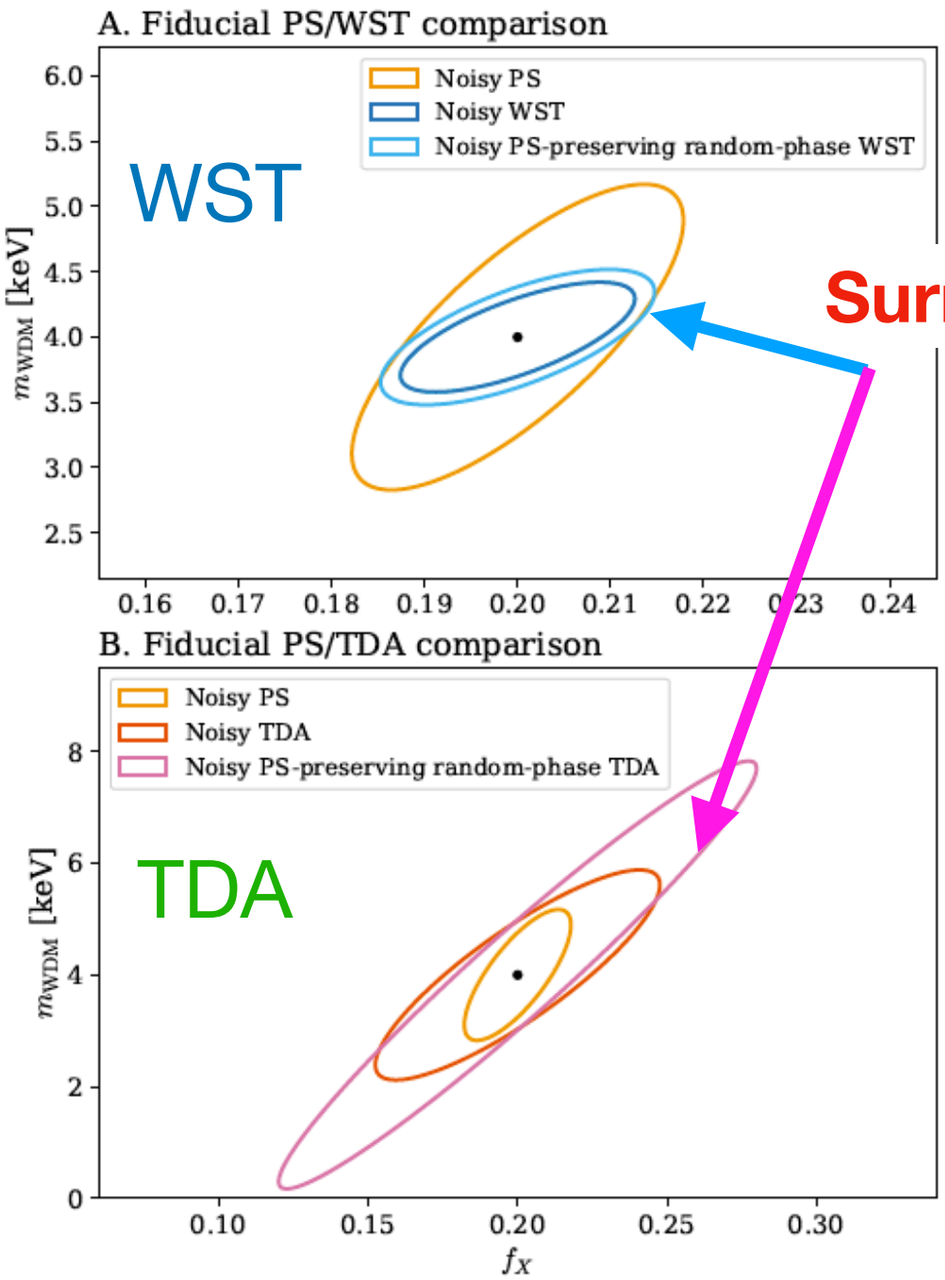
$$\eta_\phi \simeq 0.42$$

- Absolute constraining power is weaker than WST.
- The fractional phase dependence is larger than in WST.

Absolute constraining power $\neq$ fractional phase sensitivity

WST is more powerful, but most of its information remains after phase randomization, whereas TDA is weaker in absolute terms but more sensitive to phase information.

How much survives phase randomization?



Surrogate

WST

$$\sqrt{\det F} = 607.3 \rightarrow 405.1$$

$$\eta_\phi \simeq 0.33$$

- About $2/3$ of the Fisher-volume density remains after phase randomization.
- WST provides stronger constraints than PS, but most of its information survives phase randomization.

TDA

$$\sqrt{\det F} = 49.9 \rightarrow 29.2$$

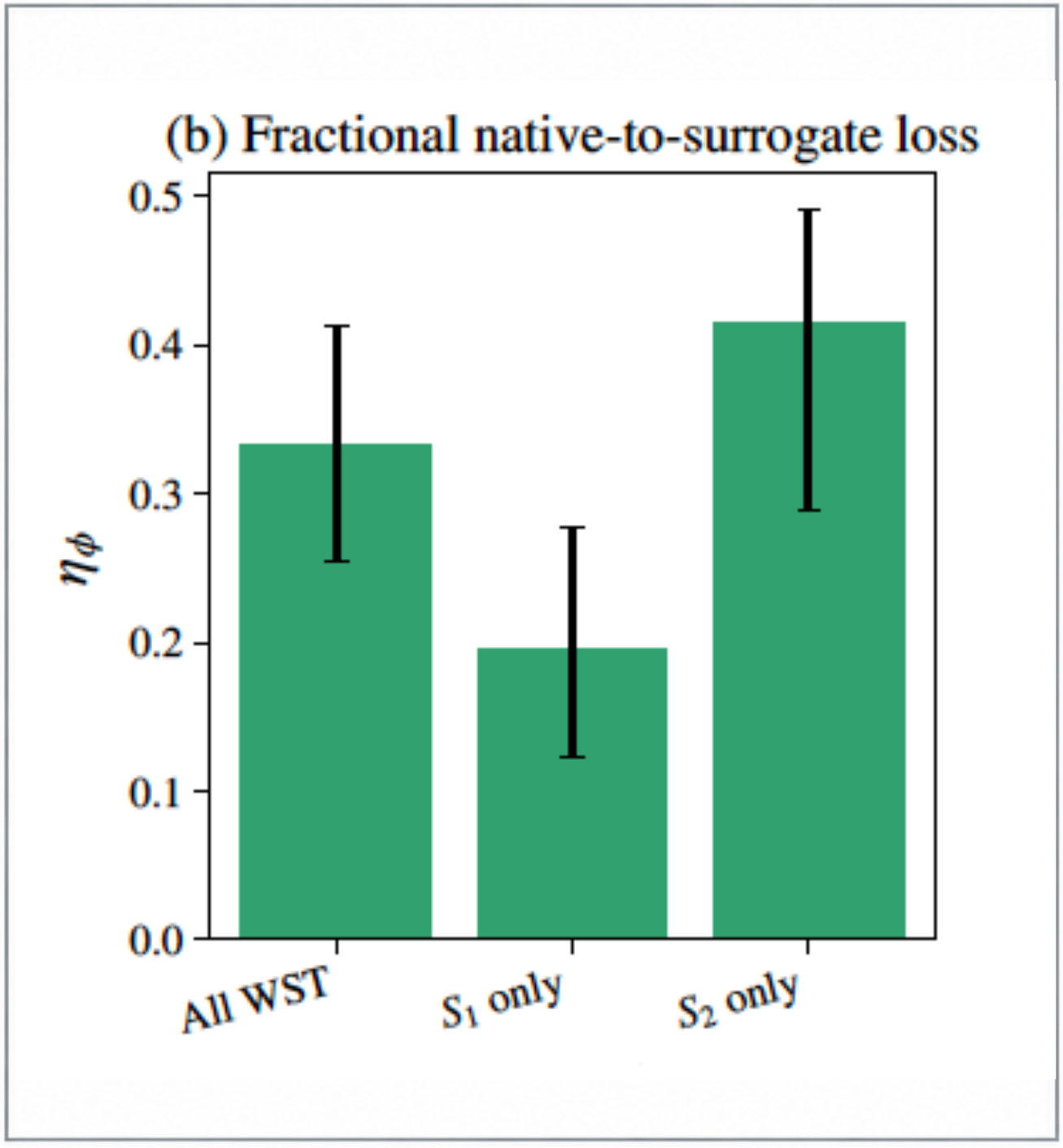
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Absolute constraining power $\neq$ fractional phase sensitivity

WST is more powerful, but most of its information remains after phase randomization, whereas TDA is weaker in absolute terms but more sensitive to phase information.

Where is the phase-sensitive information?



WST decomposition

- S_1: single-scale scattering amplitudes
- S_2: cross-scale modulation

S_1

$$\eta_\phi(S_1) \approx 0.20$$

- weaker phase sensitivity
- mainly tracks amplitude-like multiscale structure

S_2

$$\eta_\phi(S_2) \approx 0.42$$

- stronger phase sensitivity
- probes cross-scale organization of absorption features

Phase-sensitive information is concentrated more strongly in S_2.

Cross-scale modulation carries a larger relative phase-dependent contribution than S_1.

Conclusion

- **Information geometry** provides a unified way to quantify the constraining power of PS, WST, and TDA.

- **Phase randomization** reveals a **measurable contribution** from phase-sensitive, non-Gaussian structure.

- **WST** gives the strongest overall constraints, while the phase-sensitive contribution is particularly clear in the higher-order component (S_2).

Information geometry separates overall constraining power from phase-sensitive non-Gaussian information.